EEG Theta Power Varies across Roles in Mixed Human-Robot Teams

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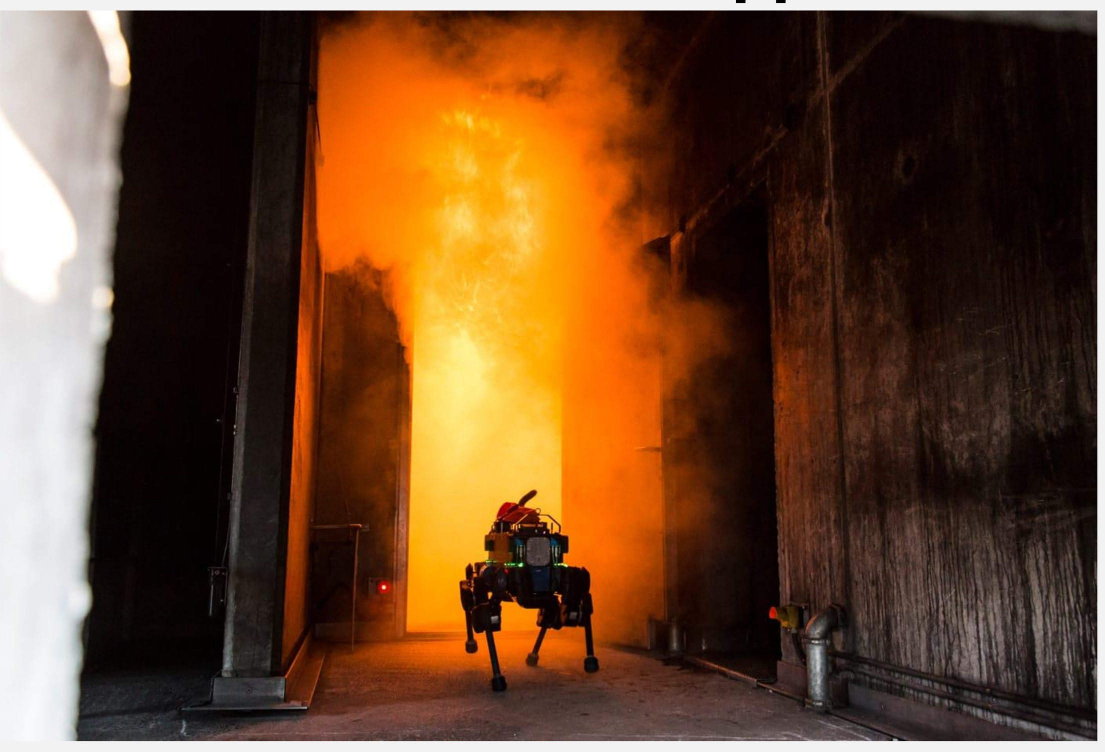
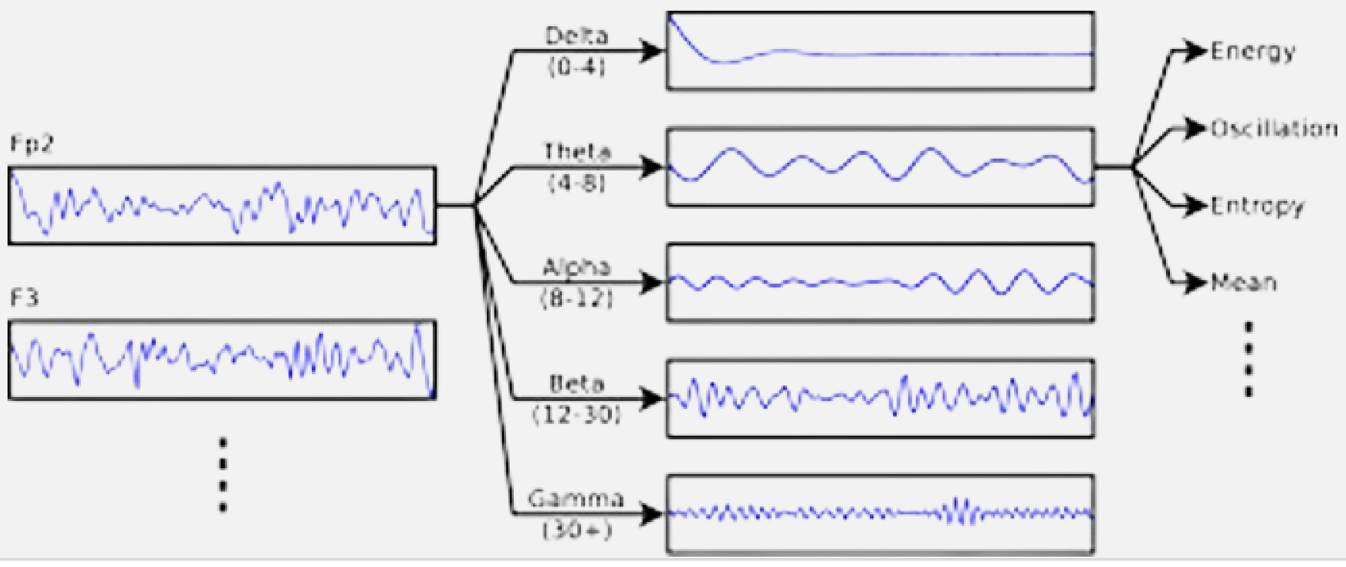
Introduction

- Search and Rescue (SAR) tasks are dangerous and **cognitively demanding tasks** [1].
  - Executive functions (planning, cognitive control, decision making, etc.)
  - Demands change based on role
- Robots should match specific demands
- EEG can help **identify optimal team attributes** and whether robots better support certain roles.
  - Frontal (FC) theta – memory, attention, & cognitive control.
  - Temporal Parietal Junction (TPJ) theta – social interactions & joint-attention [4].

ANYmal Robot [5]

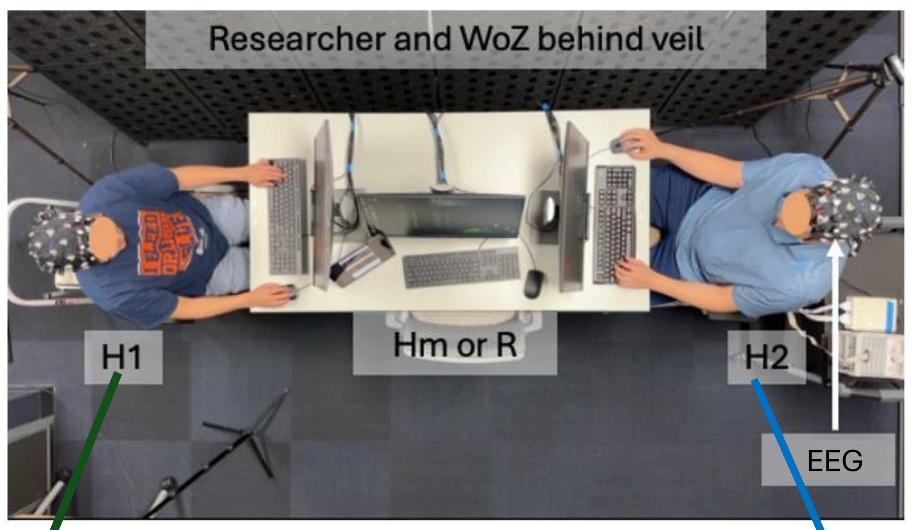




Methods

Mission: Rescue as many victims as possible while working with teammates.

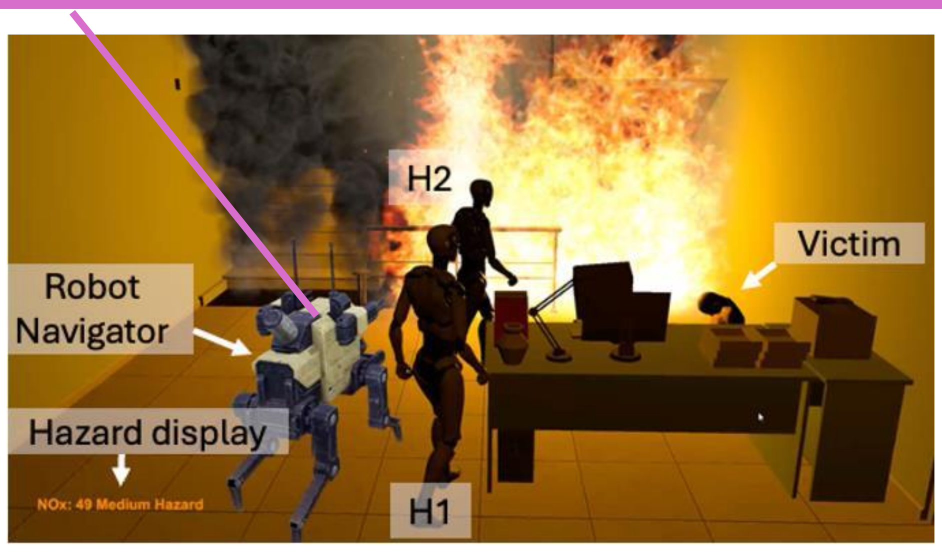
- 40 participants (16 females, 24 males),
- Five, 3-minute trials per condition:
  1. All Human (HHH)
  2. Human-Robot-Reliable (HRR)
  3. Human-Robot-Unreliable (HRU)



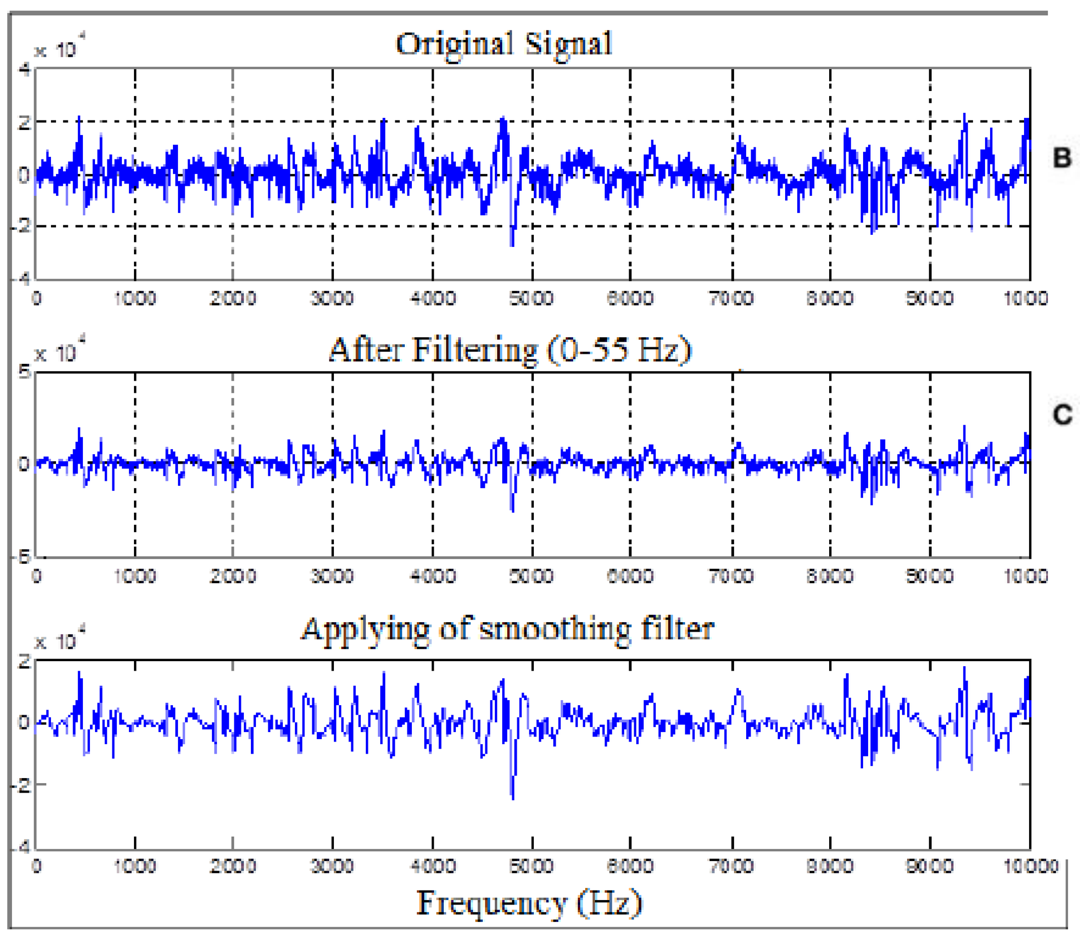
Mission Commander: Coordinates with team

Safety Officer: Monitors nitrogen oxide levels

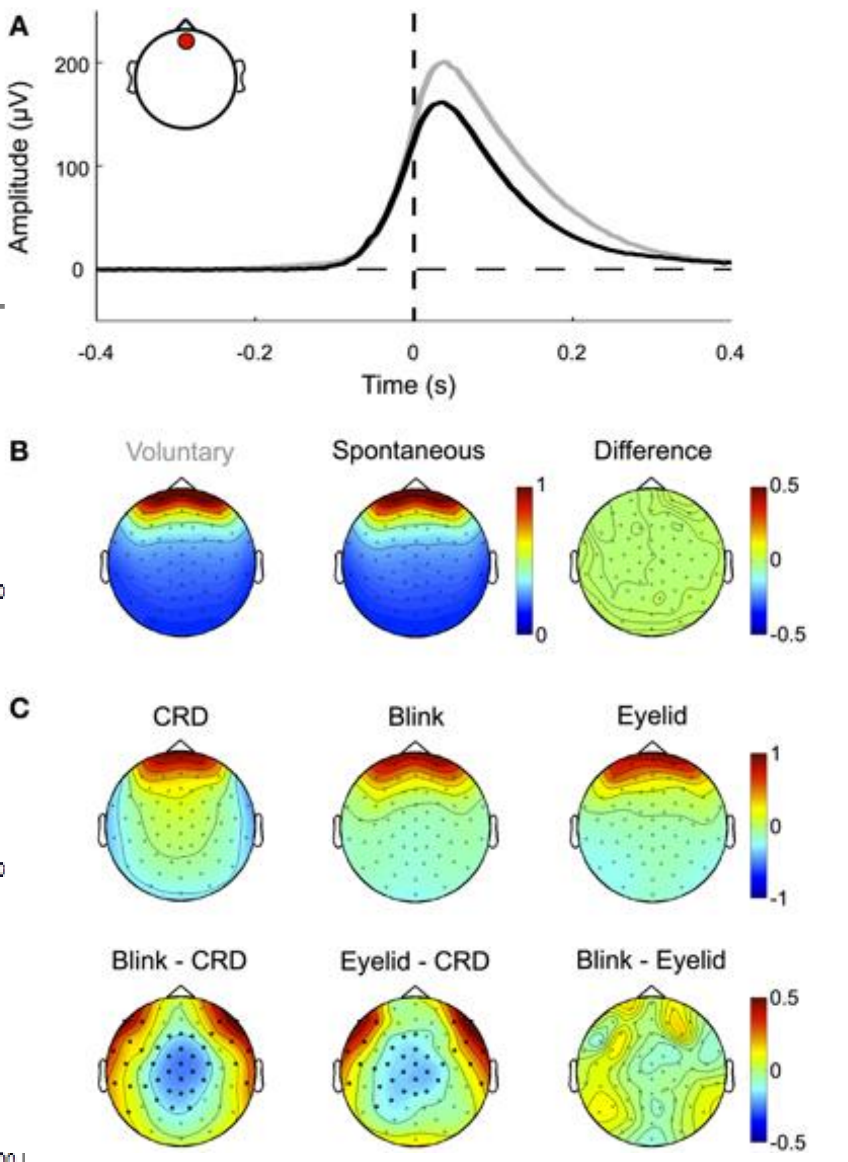
Robot or Human Navigator: Provides directions to trapped victims.



- 32-Channel EEG
- 500 Hz Sampling rate



0.1-30 Hz IIR Butterworth Filter: dampening noisy signals



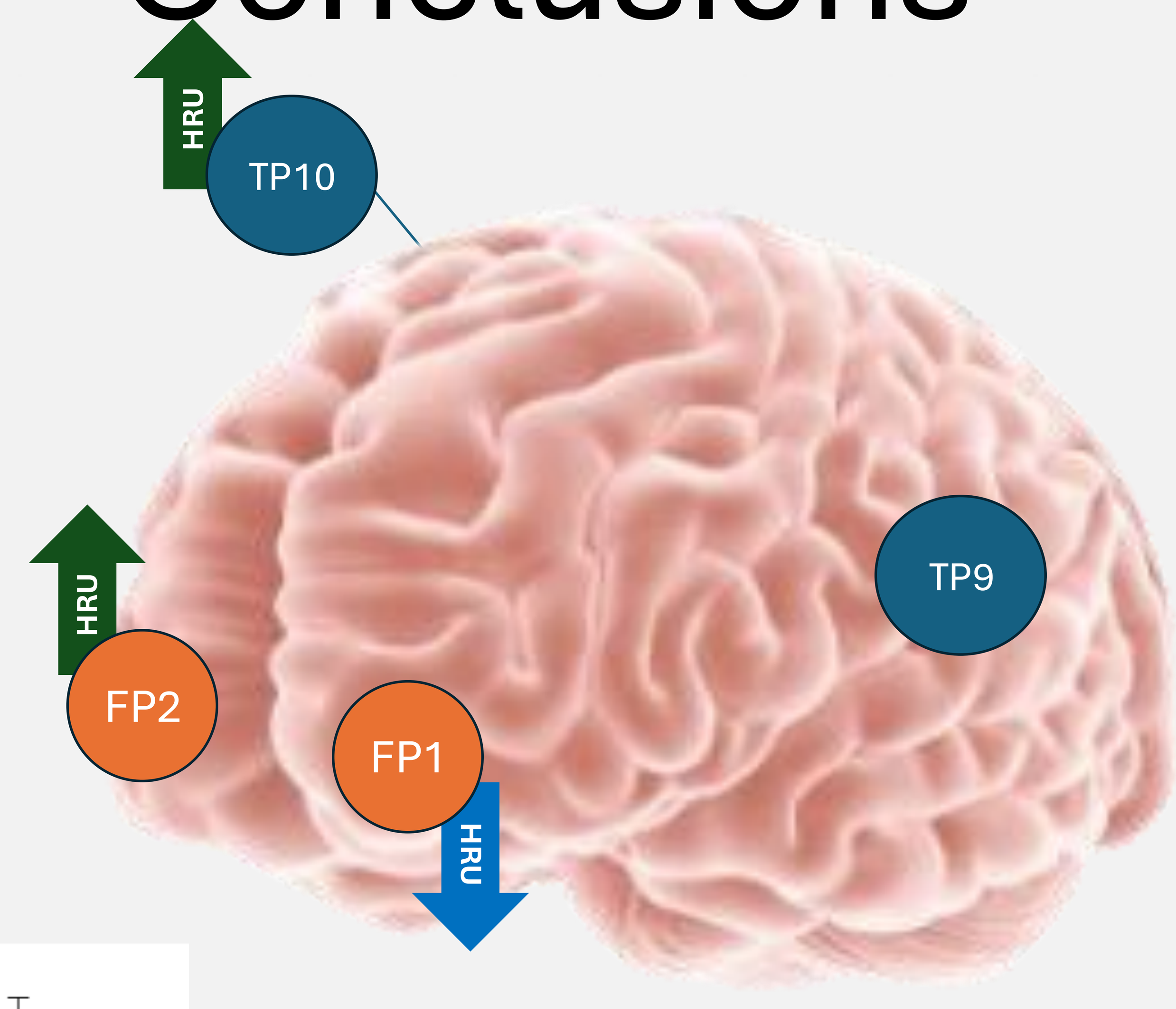
Independent Component Analysis: identify "bad" components

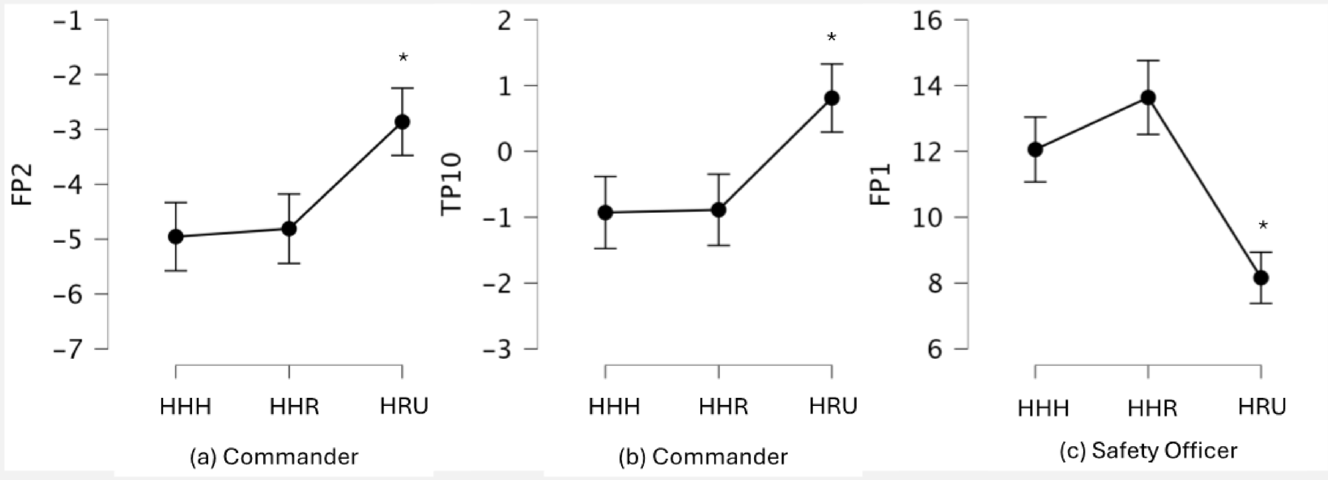
Conclusions

Mission Commander

Significantly higher theta power was observed at FP2 (p = .049) and TP10 (p = 0.05) sites in HRU condition, when compared to HHH & HRR conditions.

Mission Commanders appeared to adapt better than safety officers to changing task demands.





Safety Officer

Significantly lower theta in Fp1 for HRU conditions when compared to HHH and HRR conditions (p < .001).

Safety Officers may become less engaged in the task in HRU conditions, likely due to the mission commander's increased involvement.

Key Takeaway: Distinct Neural signatures can be measured across different team members, within different teams.

References

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